

STAT 5200, Notes 11¹

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Readings. Finish reading Chapter 6 in W.

Use of instrumental variables for the unobserved variables problem. Previously we considered the use of a proxy variable to deal with an unobserved variable. Suppose the structural model is

$$y = \mathbf{x}\beta + \gamma q + v, \tag{1}$$

where q is the unobserved variable.

Leave the unobserved variable in the error term. If the unobserved variable is put into the error term, it can cause some of the explanatory variables in $\mathbf{x}$ to be endogenous. Thus, we can use instruments for any explanatory variable that is correlated with the unobserved variable. Instrumental variables should be (i) redundant in the structural equation, (ii) uncorrelated with the unobserved variable, q , and (iii) sufficiently correlated with the endogenous variables in $\mathbf{x}$. Then 2SLS applied to (1), where the error term is taken as $\gamma q + v$, leads to consistent estimation of β .

Multiple indicators for an unobserved variable. In Section 5.3.2 Wooldridge describes an interesting methodology to deal with an unobserved variable. It requires two (or more) observable variables, termed *indicators*. Each indicator must be (i) redundant in the structural equation and (ii) correlated with the unobserved variable, and (iii) the residual in the regression of the indicator vs. the unobserved variable must be uncorrelated with the observed explanatory variables. One appends one of the indicators say q_1 , to the structural equation and then uses a second indicator, say q_2 , as an instrument for q_1 . This fits together mathematically very nicely. However, whether the assumptions needed for implementation of the methodology are valid in application is difficult to determine, and often questionable. As an example, Wooldridge considers the NLS80 data set, where one models the log of the wage and attempts to assess the return to education. The unobserved variable is ability. As indicators of ability he uses iq and kww are reversed, the estimate of the return to education is only 1.7 per cent and is not significant. Of course, the IV estimation does inflate the standard errors of the estimates (for each fit the standard error is about 2.5 times

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the standard error obtained in Wooldridge's Example 4.3, where iq is used as a proxy for ability), contributing to the non-significance. However, the point estimates are certainly not credible. One has to conclude the detailed statistical assumptions necessary to apply this methodology are not satisfied in this example.

Generated regressors. Consider a structural model with an unobserved variable, q .

$$y = \mathbf{x}\beta + \gamma q + u. \quad (2)$$

Suppose $q = f(\mathbf{w}, \delta)$, where f is known function, $\mathbf{w}$ is a vector of observed variables and δ is a vector of unknown parameters. Assume that δ can be consistently estimated, yielding $\hat{\delta}$. Then $\hat{q} = f(\mathbf{w}, \hat{\delta})$ estimates q , and $\hat{q}$ is called a *generated regressor*. What happens if we replace q by $\hat{q}$ in (2) and run an OLS regression? That is, we use

$$y = \mathbf{x}\beta + \gamma \hat{q} + u. \quad (3)$$

If we make the usual OLS assumptions about u and $(\mathbf{x}, q)$, then OLS estimation for (3) produces consistent estimation of β and γ . However, inference becomes somewhat complicated, because we have to take into account the uncertainty in the estimation of δ in forming proper standard errors. IN the special case $\gamma = 0$ there is no problem asymptotically: the standard errors one obtains from use of (3) agree asymptotically with the standard errors one would obtain from use of (2) if q were observed. Furthermore, a heteroscedasticity adjustment can be made. But if $\gamma \neq 0$, the standard errors for use of (3) need to take into account the sampling variability of $\hat{\delta}$, and standard errors for all explanatory variable coefficients in (3) are affected. In the appendix to Chapter 6, Wooldridge gives an expression which can be used to calculate appropriate standard errors. The derivation is lengthy and detailed. In view of this complication, it is wise to test the hypothesis $H_0 : \gamma = 0$ to begin with. One simply uses the usual t -statistic on $\hat{q}$ from the OLS regression with (3).

Generated instruments in 2SLS. If one is using 2SLS estimation and the instruments are generated, the story is different from that above. In fact, perhaps surprisingly, the situation is simpler and easier.

Suppose the model is $y = \mathbf{x}\beta + u$ and $\mathbf{z}$ is the vector of instrumental variables. Assume also that $\mathbf{z}$ is not observed, but that $\mathbf{z} = \mathbf{g}(\mathbf{w}, \lambda)$, where $\mathbf{g}$ consists of known functions, $\mathbf{w}$ is a vector of observed variables, and λ is estimated consistently by $\hat{\lambda}$. Then we use $\hat{\mathbf{z}} = \mathbf{g}(\mathbf{w}, \hat{\lambda})$ as the vector of instrumental variables. Then, with an appropriate assumption, the asymptotic distribution of the 2SLS estimator $\hat{\beta}$ is the same with use of λ as with the use of $\hat{\lambda}$ to construct the instruments. Furthermore, we can use standard errors assuming either homoscedasticity or heteroscedasticity for

this 2SLS estimation. One has to exercise caution, however, because this is an asymptotic result which may not be firm in some finite samples.

There is an interesting by-product of the above result: If we have a choice, it is advisable to use 2SLS with generated instruments rather than OLS with generated regressors.

Sometimes we are in a situation with both generated regressors and generated instruments. In this case, unless $\gamma = 0$ in (19), the 2SLS standard errors need to be modified because of the sampling error experienced in forming the generated regressors.

Testing for endogeneity. There have been attempts to develop tests for endogeneity. The research goes back to Durbin (1954), and it is perhaps most associated with Hausman (1978). The references are listed in W in Section 6.3.1.

Consider the model

$$y = \mathbf{x}\beta + u, \tag{4}$$

and let $\mathbf{z}$ be the vector of instruments. Let $\hat{\beta}_{2SLS}$ and $\hat{\beta}_{OLS}$ denote the 2SLS and OLS estimators of the parameter vector. The test for endogeneity is derived under the null hypothesis that *all* the variables in $\mathbf{x}$ are exogenous. Under this hypothesis, the two estimators should differ only by sampling error. A test can be based upon the difference $\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}$. The asymptotic distribution of

$$\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) \tag{5}$$

is normal, and thus a type of Wald statistic or F -statistic can be formed to test the null hypothesis. Note that the excluded instruments in $\mathbf{z}$ are used to form the 2SLS estimator of the parameter vector. Rejection of the null hypothesis will suggest that at least one of the elements of $\mathbf{x}$ allowed to be endogenous under the alternative hypothesis (at least one of the elements for which we have instrumented) is in fact endogenous.

To form the test statistic we need to determine the asymptotic covariance matrix of (5) *under the null hypothesis that all the elements of $\mathbf{x}$ are exogenous*. Assume Wooldridge's Assumptions 2.SLS.1–2.SLS.3 hold with $\mathbf{w}$, where $\mathbf{w}$ consists of all the elements of $\mathbf{x}$ and all the excluded instruments in $\mathbf{z}$. These assumptions include homoscedasticity (that u is measured with constant variance σ^2), specification of the null hypothesis, and exogeneity of the excluded instruments. This is part a of Problem 6.12 in W.

Recall that $\mathbf{x}^*$ denotes the projection of the $\mathbf{x}$ vector onto the vector of instruments $\mathbf{z}$. It follows that

$$\mathbf{x} = \mathbf{x}^* + \mathbf{r}, \quad E[\mathbf{x}^{*'} \mathbf{r}] = \mathbf{0}, \quad E[\mathbf{x}^{*'} \mathbf{x}] = E[\mathbf{x}' \mathbf{x}^*] = E[\mathbf{x}^{*'} \mathbf{x}^*]. \quad (6)$$

By Theorem 4.2 in W,

$$\sqrt{N}(\hat{\beta}_{OLS} - \beta) = (E[\mathbf{x}' \mathbf{x}])^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i + o_p(1) \quad (7)$$

converges in distribution, as N tends to infinity, to

$$N \left(\mathbf{0}, \sigma^2 (E[\mathbf{x}' \mathbf{x}])^{-1} \right) \quad (8)$$

Similarly, by Theorem 5.2 in W,

$$\sqrt{N}(\hat{\beta}_{2SLS} - \beta) = (E[\mathbf{x}^{*'} \mathbf{x}^*])^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}^{*'}_i u_i + o_p(1) \quad (9)$$

converges in distribution, as N tends to infinity, to

$$N \left(\mathbf{0}, \sigma^2 (E[\mathbf{x}^{*'} \mathbf{x}^*])^{-1} \right). \quad (10)$$

Next, the asymptotic covariance matrix of (5) is

$$\begin{aligned} Avar \left(\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) \right) &= Avar \left(\sqrt{N}(\hat{\beta}_{2SLS} - \beta) - \sqrt{N}(\hat{\beta}_{OLS} - \beta) \right) \\ &= Avar \left(\sqrt{N}(\hat{\beta}_{2SLS} - \beta) \right) \\ &\quad - Acov \left(\sqrt{N}(\hat{\beta}_{2SLS} - \beta), \sqrt{N}(\hat{\beta}_{OLS} - \beta) \right) \\ &\quad - Acov \left(\sqrt{N}(\hat{\beta}_{OLS} - \beta), \sqrt{N}(\hat{\beta}_{2SLS} - \beta) \right) \\ &\quad + Avar \left(\sqrt{N}(\hat{\beta}_{OLS} - \beta) \right). \end{aligned} \quad (11)$$

The evaluation of the second and fifth lines of (11) is shown in (10) and (8), respectively. The negative of the third line of (11) is, from (7) and (9),

$$\begin{aligned} & \left(E[\mathbf{x}^{*'} \mathbf{x}^*] \right)^{-1} E \left[\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}^{*'}_i u_i \frac{1}{\sqrt{N}} \sum_{j=1}^N \mathbf{x}_j u_j \right] (E[\mathbf{x}' \mathbf{x}])^{-1} \\ &= \left(E[\mathbf{x}^{*'} \mathbf{x}^*] \right)^{-1} \left[\frac{1}{N} \sum_{i=1}^N E[\mathbf{x}^{*'}_i u_i^2 \mathbf{x}_i] \right] (E[\mathbf{x}' \mathbf{x}])^{-1} \\ &= \left(E[\mathbf{x}^{*'} \mathbf{x}^*] \right)^{-1} \left[\sigma^2 E[\mathbf{x}^{*'}_i \mathbf{x}_i] \right] (E[\mathbf{x}' \mathbf{x}])^{-1} \quad (\text{by homoscedasticity}) \\ &= \sigma^2 \left(E[\mathbf{x}^{*'} \mathbf{x}^*] \right)^{-1} E[\mathbf{x}^{*'}_i \mathbf{x}_i] (E[\mathbf{x}' \mathbf{x}])^{-1} \quad (\text{by (6)}) \\ &= \sigma^2 (E[\mathbf{x}' \mathbf{x}])^{-1}. \end{aligned} \quad (12)$$

The negative of the fourth line of (11) is also equal to $\sigma^2 (E[\mathbf{x}'\mathbf{x}])^{-1}$. It follows that (11) is

$$Avar\left(\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})\right) = \sigma^2 \left(E[\mathbf{x}^*{}'\mathbf{x}^*]\right)^{-1} - \sigma^2 (E[\mathbf{x}'\mathbf{x}])^{-1}. \quad (13)$$

The matrix (13) is positive semidefinite, because it is a covariance matrix.

However, its rank is equal to the number of elements of $\mathbf{x}$ allowed to be endogenous under the alternative hypothesis, and thus it is not positive definite and therefore not invertible. (I'm working here with the full β vector, including the intercept, and, of course, the constant element of $\mathbf{x}$ is uncorrelated with u , and is thus necessarily exogenous. That is, with the above development, all of the elements of $\mathbf{x}$ cannot be allowed to be endogenous under the alternative hypothesis. We could work with a subset of the full β vector and develop the analogue of (13) for the subset, and then it would be possible for all the corresponding explanatory variables in the subset to be allowed to be endogenous under the alternative hypothesis.) Let's prove that the rank of (13) is equal to the number of explanatory variables allowed to be endogenous under the alternative hypothesis. We consider

$$\left(E[\mathbf{x}^*{}'\mathbf{x}^*]\right)^{-1} - (E[\mathbf{x}'\mathbf{x}])^{-1}. \quad (14)$$

In (6), write $\mathbf{x} = \mathbf{x}^* + \mathbf{r}$ as $(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^*, \mathbf{x}_2^*) + (\mathbf{r}_1, \mathbf{r}_2)$, where the number of elements in $\mathbf{x}_1$, $\mathbf{x}_1^*$, and $\mathbf{r}_1$ is the number of exogenous variables in $\mathbf{x}$ (say, K_1), and the number of elements in $\mathbf{x}_2$, $\mathbf{x}_2^*$, and $\mathbf{r}_2$ is the number of endogenous variables in $\mathbf{x}$ (say, K_2). But, because $\mathbf{x}_1$ consists of exogenous variables, $\mathbf{x}_1^* = \mathbf{x}_1$ and $\mathbf{r}_1 = \mathbf{0}$. That is, $(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1, \mathbf{x}_2^* + \mathbf{r}_2)$. Further, note that

$$E[\mathbf{x}_1'\mathbf{r}_2] = \mathbf{0} \quad \text{and} \quad E[\mathbf{x}_2^*{}'\mathbf{r}_2] = \mathbf{0}.$$

Thus we can write

$$\left[E[\mathbf{x}^*{}'\mathbf{x}^*]\right]^{-1} = \begin{pmatrix} E[\mathbf{x}_1'\mathbf{x}_1] & E[\mathbf{x}_1'\mathbf{x}_2^*] \\ E[\mathbf{x}_2^*{}'\mathbf{x}_1] & E[\mathbf{x}_2^*{}'\mathbf{x}_2^*] \end{pmatrix}^{-1}$$

and Thus we can write

$$\left[E[\mathbf{x}'\mathbf{x}]\right]^{-1} = \begin{pmatrix} E[\mathbf{x}_1'\mathbf{x}_1] & E[\mathbf{x}_1'\mathbf{x}_2^*] \\ E[\mathbf{x}_2^*{}'\mathbf{x}_1] & E[\mathbf{x}_2^*{}'\mathbf{x}_2^*] + E[\mathbf{r}_2'\mathbf{r}_2] \end{pmatrix}^{-1}.$$

For ease of writing, we reexpress this as

$$\left[E[\mathbf{x}^*{}'\mathbf{x}^*]\right]^{-1} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}^{-1} \quad \text{and} \quad \left[E[\mathbf{x}'\mathbf{x}]\right]^{-1} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} + \mathbf{B}_{22} \end{pmatrix}^{-1},$$

where the correspondence is obvious. Suppose there are K elements in $\mathbf{x}$, with K_1 of them exogenous and K_2 of them endogenous (and $K = K_1 + K_2$, of course, so that the blocks in the matrices are $K_1 \times K_1$, $K_1 \times K_2$, $K_2 \times K_1$, and $K_2 \times K_2$). With this notation, (14) is

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}^{-1} - \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} + \mathbf{B}_{22} \end{pmatrix}^{-1}. \quad (15)$$

We consider Woodbury's matrix identity. It gives the inverse of a rank k adjustment of a given rank n matrix, $n > k$, as a rank k adjustment of the inverse of the original matrix. We state it as a lemma.

Lemma. Let $\mathbf{A}$ be $n \times n$, $\mathbf{U}$ be $n \times k$, $\mathbf{V}$ be $k \times n$, and $\mathbf{B}$ be $k \times k$. Then

$$(\mathbf{A} + \mathbf{UBV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{B}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}, \quad (16)$$

where all the inverses are assumed to exist.

Proof. Consider the system

$$\begin{pmatrix} \mathbf{A} & \mathbf{U} \\ \mathbf{V} & -\mathbf{B}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} \mathbf{I} \\ \mathbf{0} \end{pmatrix}.$$

Expanding, we obtain

$$(i) \quad \mathbf{AX} + \mathbf{UY} = \mathbf{I},$$

$$(ii) \quad \mathbf{VX} - \mathbf{B}^{-1}\mathbf{Y} = \mathbf{0}.$$

From (i), $\mathbf{X} = \mathbf{A}^{-1}(\mathbf{I} - \mathbf{UY})$. Substitution of this into (ii) yields

$$\mathbf{VA}^{-1}(\mathbf{I} - \mathbf{UY}) - \mathbf{B}^{-1}\mathbf{Y} = \mathbf{0}, \quad \text{or} \quad (\mathbf{B}^{-1} + \mathbf{VA}^{-1}\mathbf{U})\mathbf{Y} = \mathbf{VA}^{-1}.$$

$$(iii) \quad \mathbf{Y} = (\mathbf{B}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}.$$

or

$$\mathbf{X} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{B}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}. \quad (17)$$

Now, (ii) gives $\mathbf{Y} = \mathbf{BVX}$. Inserting this into (i), we have $\mathbf{AX} + \mathbf{UBVX} = \mathbf{I}$, or $\mathbf{X} = (\mathbf{A} + \mathbf{UBV})^{-1}$. This and (17) give (16). $\square$

To find the rank of (15), we apply the lemma with

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad (K_1 + K_2) \times (K_1 + K_2); \quad \mathbf{B} = \mathbf{B}_{22}, \quad K_2 \times K_2;$$

$$\mathbf{U} = \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_{K_2} \end{pmatrix}, \quad (K_1 + K_2) \times K_2; \quad \mathbf{V} = (\mathbf{0} \quad \mathbf{I}_{K_2}), \quad K_2 \times (K_1 + K_2).$$

Then (15) is $\mathbf{A}^{-1} - (\mathbf{A} + \mathbf{UBV})^{-1}$. By the lemma this is

$$\begin{aligned} & \mathbf{A}^{-1} - \mathbf{A}^{-1} + \mathbf{A}^{-1} \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_{K_2} \end{pmatrix} \left(\mathbf{B}^{-1} + (\mathbf{0} \quad \mathbf{I}_{K_2}) \mathbf{A}^{-1} \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_{K_2} \end{pmatrix} \right)^{-1} (\mathbf{0} \quad \mathbf{I}_{K_2}) \mathbf{A}^{-1} \\ & = \mathbf{A}^{-1} \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_{K_2} \end{pmatrix} \left(\mathbf{B}^{-1} + (\mathbf{0} \quad \mathbf{I}_{K_2}) \mathbf{A}^{-1} \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_{K_2} \end{pmatrix} \right)^{-1} (\mathbf{0} \quad \mathbf{I}_{K_2}) \mathbf{A}^{-1} \end{aligned}$$

a matrix which has rank K_2 .

To test the null hypothesis that all the elements of $\mathbf{x}$ are exogenous, we can use the Wald-like statistic

$$\left(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}\right)' \left((\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1} - (\mathbf{X}'\mathbf{X})^{-1}\right)^- \left(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}\right) / \hat{\sigma}^2, \quad (18)$$

where a generalized inverse of the asymptotic covariance matrix is employed. In (18) the first-stage fitted values $\hat{\mathbf{x}}_i$ are used in place of the population projections $\mathbf{x}_i^*$. See (13). The estimate $\hat{\sigma}^2$ can be obtained from the 2SLS residuals, or from the OLS residuals, or from both sets of residuals. The asymptotic distribution under the null hypothesis of the statistic (18) is the same for any choice of generalized inverse, and it is a chi-square distribution. However, it is awkward to use the statistic because of the need to calculate a generalized inverse. Fortunately, Hausman has proved that there is a regression-based test of endogeneity which is asymptotically equivalent to the test based on (18), and this regression-based test is what is commonly used. Recall that, if all the elements of $\mathbf{x}$ we are considering are endogenous (under the alternative hypothesis), the required asymptotic covariance matrix does have full rank.

Before briefly presenting the regression-based test, let's note that the statistic (18) assumes homoscedasticity. It is straightforward to modify the derivation of (18) to produce a statistic that is robust to heteroscedasticity. For example, use (9) in Notes 4, or modifications of it as in the sandwich package in R. This is part b of Problem 6.12 in W.

The regression-based test of endogeneity is easy to implement. For each potentially endogenous explanatory variable, we run the first-stage regression in 2SLS of the variable vs. the vector of instruments, and we calculate the residuals from this regression. Then we run a regression with each of the elements of $\mathbf{x}$ and the vector of first-stage residuals as explanatory variables. The test of endogeneity is then a standard Wald test or F -test. This procedure can be made robust to heteroscedasticity.

Let's illustrate this using Wooldridge's notation (page 132). Suppose the population model is

$$y_1 = \mathbf{z}_1\delta_1 + \mathbf{y}_2\alpha_1 + u_1, \quad (19)$$

where $(\mathbf{z}_1, y_2)$ is the vector of explanatory variables, the elements of $\mathbf{z}_1$ are assumed to be exogenous, and the elements of $\mathbf{y}_2$ are potentially endogenous. The parameter vector α_1 is $G_1 \times 1$. Further, $\mathbf{z}$ is the vector of instrumental variables. The first-stage regressions are

$$\mathbf{y}_2 = \mathbf{z}\Pi_2 + \mathbf{v}_2, \quad (20)$$

and $\hat{\mathbf{v}}_2$ denotes the residual vector from performance of the (first stage) regressions (20). Estimate the regression

$$y_1 = \mathbf{z}_1\delta_1 + \mathbf{y}_2\alpha_1 + \hat{\mathbf{v}}_2\rho_1 + error. \quad (21)$$

The regression-based test of endogeneity is simply a Wald test or an F -test of the null hypothesis

$$H_0 : \rho_1 = \mathbf{0}. \quad (22)$$

This test is made robust to heteroscedasticity in the usual way, with (9) in Notes 4 or the sandwich package in R.

The rationale for this procedure is straightforward. Given (19), we want to test the null hypothesis

$$H_0 : \text{cov}(\mathbf{y}_2, u_1) = \mathbf{0}, \quad (23)$$

That is, that the elements of $\mathbf{y}_2$ are exogenous. From the first-stage regressions (20), the covariance in (23) is

$$\text{cov}(\mathbf{y}_2, u_1) = \text{cov}(\mathbf{z}\Pi_2 + \mathbf{v}_2 u_1) = \text{cov}(\mathbf{v}_2, u_1) = E[\mathbf{v}_2 u_1],$$

because $\mathbf{z}$ is uncorrelated with u_1 and u_1 has mean 0. The linear projection of u_1 on $\mathbf{v}_2$ is

$$u_1 = \mathbf{v}_2 \rho_1 + e_1. \quad (24)$$

Therefore H_0 is equivalent to

$$H_0 : \rho_1 = \mathbf{0}. \quad (25)$$

Substitute (24) into (19), yielding

$$y_1 = \mathbf{z}_1 \delta_1 + \mathbf{y}_2 \alpha_1 + \mathbf{v}_2 \rho_1 + e_1. \quad (26)$$

Use the residuals from the first-stage regression (20), $\hat{\mathbf{v}}_2$, in (26) in place of $\mathbf{v}_2$. That is, run the regression

$$y_1 = \mathbf{z}_1 \delta_1 + \mathbf{y}_2 \alpha_1 + \hat{\mathbf{v}}_2 \rho_1 + e_1, \quad (27)$$

and test the hypothesis (25) with a Wald test or an F -test. In performing this test, it is not necessary to rescale the standard errors of $\hat{\rho}_1$ as is done with two-stage least squares estimation, because $\hat{\mathbf{v}}_2$ is a generated regressor, and adjustment is not needed under the null hypothesis (22).

Testing overidentifying restriction. In Section 6.3.2 Wooldridge discusses testing overidentifying restrictions. It is common to use more instruments than are needed to identify a model. One may want to test whether the additional instruments are in fact uncorrelated with the error term. A suggestion by Hausman is to compare the 2SLS estimator using all the instruments to the 2SLS estimator using just enough instruments to identify the model. If all the instruments are valid, these two should differ only by sampling error. As with the endogeneity test associated with Hausman, one can derive a quadratic form in the coefficient estimator differences. This expression is difficult to work with. There is a regression-based test due to Sargan (1958) which is easy to sue and is commonly employed.

Assume the model is given by (19), where $\mathbf{z}_1$ is $1 \times L_1$ and $\mathbf{y}_2$ is $1 \times G_1$. The elements of $\mathbf{z}_1$ are assumed to be exogenous and those of $\mathbf{y}_2$ to be endogenous. The $1 \times L$ vector of all the instruments is $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2)$, where $\mathbf{z}_2$ is $1 \times L_2$. Thus, $L = L_1 + L_2$, and, because of the overidentification, $L_2 > G_1$. Sargan's test procedure is as follows. Perform the regression (19) by 2SLS using *all* the instruments, and form the 2SLS residuals, $\hat{u}_1$. Then regress $\hat{u}_1$ on $\mathbf{z}$, including a constant, with OLS. Under the null hypothesis

$$H_0 : E[\mathbf{z}'u_1] = \mathbf{0},$$

and homoscedasticity, the Lagrange multiplier statistic NR_u^2 is asymptotically chi-square with $L_2 - G_1$ degrees of freedom. Note that $L_2 - G_1$ is number of overidentifying restrictions. It is easy to modify this procedure to make it robust to heteroscedasticity. The details are given on page 135 of Wooldridge.

There are some problems with this test. If we reject the null hypothesis, the test doesn't indicate which instruments are not exogenous. And, if we do not reject the null hypothesis, there is no guarantee that the instruments are not endogenous. For example, there can be biased attributable to the instruments which are similar with a full set and with a reduced set of instruments, leading to cancellation in the test statistic. Or there could be misspecification of the model altogether. Furthermore, 2SLS standard errors tend to be large, and for this reason alone we might fail to reject the null hypothesis. The same problem associated with large 2SLS standard errors is potentially present for the Hausman test of endogeneity described in the previous section.

Testing for heteroscedasticity. Although the presence of heteroscedasticity does not affect the consistency of OLS and 2SLS estimators, it does affect the construction of hypothesis tests and confidence intervals. Let's consider a framework to test for the presence of heteroscedasticity. Assume a model where all the explanatory variables are exogenous,

$$y = \beta_0 + \mathbf{x}\beta + u, \quad \text{with } E[u|\mathbf{x}] = 0. \quad (28)$$

The zero conditional mean assumption is needed for the heteroscedasticity test specified below. With this assumption, the null hypothesis of homoscedasticity is

$$H_0 : E[u^2|\mathbf{x}] = \sigma^2. \quad (29)$$

Under this hypothesis, it follows that $E[u^2] = \sigma^2$.

The tests of the null hypothesis essentially consider covariances such as

$$\text{cov}(\mathbf{h}(\mathbf{x}), u^2), \quad (30)$$

where $\mathbf{h}(\mathbf{x})$ is a $1 \times Q$ vector of functions of $\mathbf{x}$. Under the null hypothesis, we have

$$\begin{aligned} \text{cov}(\mathbf{h}(\mathbf{x}), u^2) &= E[\mathbf{h}(\mathbf{x})u^2] - \sigma^2 E[\mathbf{h}(\mathbf{x})] \\ &= E[E[\mathbf{h}(\mathbf{x})u^2 | \mathbf{x}]] - \sigma^2 E[\mathbf{h}(\mathbf{x})] \\ &= E[\mathbf{h}(\mathbf{x})E[u^2 | \mathbf{x}]] - \sigma^2 E[\mathbf{h}(\mathbf{x})] \\ &= \sigma^2 E[\mathbf{h}(\mathbf{x})] - \sigma^2 E[\mathbf{h}(\mathbf{x})] \\ &= \mathbf{0}'. \end{aligned}$$

Therefore, we can use a regression to test H_0 . Write the regression model

$$u_i^2 = \delta_0 + \mathbf{h}(\mathbf{x}_i)\delta + v_i. \quad (31)$$

We assume that $\text{cov}(\mathbf{h}(\mathbf{x}))$ is a matrix of rank Q , to avoid perfect collinearity. It follows that, under the null hypothesis (29), we must have

$$E[v_i | \mathbf{x}_i] = 0, \quad \delta = \mathbf{0}, \quad \delta_0 = \sigma^2. \quad (32)$$

Therefore, one tests the null hypothesis (29) by testing that $\delta = \mathbf{0}$ with a Wald test or an F -test. To do so, we replace the unobserved u_i^2 in (31) by $\hat{u}_i^2$, where the $\hat{u}_i$ are the residuals from fitting the regression $y = \beta_0 + \mathbf{x}\beta + u$. Note that, under the null hypothesis (29), $v_i \geq -\sigma^2$. Also, note that, under H_0 , the standard Wald test or the standard F -test under the assumption of homoscedasticity of the error term v_i in (31) requires that

$$E[v_i^2 | \mathbf{x}_i] \text{ is constant,}$$

which in turn requires that

$$E[(u_i^2 - \sigma^2)^2 | \mathbf{x}_i] \text{ is constant,}$$

hence that

$$E[u_i^4 | \mathbf{x}_i] = \kappa_4 + 3\sigma^4, \quad (33)$$

where κ_4 is the fourth-order cumulant of u_i . Condition (33) is called *homokurtosis*.

As noted, we don't observe the u_i^2 values, and, thus, to run the regression (31), we replace u_i^2 with $\hat{u}_i^2$, the squared residuals from the OLS regression in (28). It can be shown that this replacement leads to the correct asymptotics for the test of the null hypothesis (29). The details are in Section 6.3.4 and Problem 6.5 in W (see below).

One has to choose the elements in $\mathbf{h}(\mathbf{x})$. A version of the Breusch-Pagan test uses $\mathbf{h}(\mathbf{x}) = \mathbf{x}$ (recall that, for this discussion, $\mathbf{x}$ does not include the constant 1). The test can be performed in R by the function `ncvTest()` in the `car` package, and by the function `bptest()` in the `lmtest` package. White's test chooses $\mathbf{h}(\mathbf{x})$ to include all the elements of $\mathbf{x}$ and all squares and products of the elements of $\mathbf{x}$. The R functions in the previous sentence can be used. Unless the dimension of $\mathbf{h}(\mathbf{x})$

is very small, the number of numerator degrees of freedom of the F -test can be quite large. A test which has only two numerator degrees of freedom uses $\mathbf{h}(\mathbf{x}_i, \hat{\beta}) = (\hat{y}_i, \hat{y}_i^2)$, where the $\hat{y}_i = \mathbf{x}_i' \hat{\beta}$ are the OLS fitted values, in place of $\mathbf{h}(\mathbf{x}_i)$. Wooldridge describes the justification for this test in his discussion starting at the bottom of page 140. The justification includes consideration of generated regressors. One can give a test of heteroscedasticity if some elements of $\mathbf{x}$ are endogenous, and there are some added complications. One uses $\mathbf{h}(\mathbf{z}_i)$ instead of $\mathbf{h}(\mathbf{x}_i)$. See the discussion on page 141 in W.

Problem 6.5 in W.

a. Consider the model $y = \beta_0 + \mathbf{x}\beta + u$, where $E[u|\mathbf{x}] = 0$ and $E[u^2|\mathbf{x}] = \sigma^2$. By iterated expectations, these imply

$$E[u^2] = \text{var}(u) = \sigma^2.$$

We need to verify (6.39) in W,

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{h}_i'(\hat{u}_i^2 - \hat{\sigma}^2) = \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(u_i^2 - \sigma^2) + o_p(1). \quad (6.39)$$

Here $\mathbf{h}_i \equiv \mathbf{h}(\mathbf{x}_i)$ is a $1 \times Q$ vector of functions of $\mathbf{x}_i$, $\mu_h = E[\mathbf{h}_i]$, $\hat{u}_i$ is the i th residual from OLS estimation of β , and

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N \hat{u}_i^2 \quad (34)$$

is the estimator of σ^2 . For this problem, it is okay to divide by N in (34) instead of the degrees of freedom $N - K$, because we are concerned only with asymptotic behavior.

First, let's note we can write the left-hand side of (6.39) as

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(\hat{u}_i^2 - \hat{\sigma}^2), \quad (35)$$

because

$$\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2) = \sum_{i=1}^N \left(\hat{u}_i^2 - N^{-1} \sum_{j=1}^N \hat{u}_j^2 \right) = \sum_{i=1}^N \hat{u}_i^2 - \sum_{j=1}^N \hat{u}_j^2 = 0.$$

Expression (35) is

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(\hat{u}_i^2 - \sigma^2) - \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(\hat{\sigma}^2 - \sigma^2). \quad (36)$$

Consider the second summation in (36). By the central limit theorem, the factor

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'$$

is $O_p(1)$, because it converges in distribution to a normal random vector. We have that $\hat{\sigma}^2$ converges in probability to σ^2 , so that $\hat{\sigma}^2 - \sigma^2 = o_p(1)$. Thus, the second summation in (36) is $O_p(1)o_p(1) = o_p(1)$, so that (36) is

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' (\hat{u}_i^2 - \sigma^2) + o_p(1). \quad (37)$$

Recall that $\hat{u}_i^2 = u_i^2 - 2u_i \mathbf{x}_i'(\hat{\beta} - \beta) + \mathbf{x}_i'(\hat{\beta} - \beta)(\hat{\beta} - \beta)' \mathbf{x}_i'$. Then (37) is

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' (u_i^2 - \sigma^2) - \frac{2}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' u_i \mathbf{x}_i' (\hat{\beta} - \beta) + \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' \mathbf{x}_i' (\hat{\beta} - \beta) (\hat{\beta} - \beta)' \mathbf{x}_i' + o_p(1). \quad (38)$$

Consider the second summation in (38). We first note that

$$\begin{aligned} E[(\mathbf{h}_i - \mu_h)' u_i \mathbf{x}_i] &= E[E[(\mathbf{h}_i - \mu_h)' u_i \mathbf{x}_i \mid \mathbf{x}_i]] \\ &= E[(\mathbf{h}_i - \mu_h)' \mathbf{x}_i E[u_i \mid \mathbf{x}_i]] \\ &= E[(\mathbf{h}_i - \mu_h)' \mathbf{x}_i \cdot 0] \\ &= \mathbf{0}, \end{aligned} \quad (39)$$

a matrix of zero. By (39) and the central limit theorem, the random vector

$$\text{vec} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' u_i \mathbf{x}_i \right) = \frac{1}{\sqrt{N}} \sum_{i=1}^N u_i (\mathbf{x}_i' \otimes (\mathbf{h}_i - \mu_h)')$$

[use $\text{vec}(\mathbf{DEF}) = (\mathbf{F}' \otimes \mathbf{D})\text{vec}(\mathbf{E})$; see Appendix for a discussion] converges in distribution to a normal random vector with mean vector $\mathbf{0}$ and covariance matrix

$$\begin{aligned} E[u_i^2 (\mathbf{x}_i' \otimes (\mathbf{h}_i - \mu_h)') (\mathbf{x}_i \otimes (\mathbf{h}_i - \mu_h))] &= E[u_i^2 (\mathbf{x}_i' \mathbf{x}_i \otimes (\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h))] \\ &= \sigma^2 E[\mathbf{x}_i' \mathbf{x}_i \otimes (\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h)]. \end{aligned} \quad (40)$$

where we assume that the expectations in (40) exist. It follows that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' u_i \mathbf{x}_i = O_p(1).$$

Moreover, $\hat{\beta} - \beta = O_p(N^{-1/2})$. Therefore, the second summation in (38) is $O_p(1) \cdot O_p(N^{-1/2}) = o_p(1)$.

The vec of the third summation in (38) is

$$\begin{aligned} & \frac{1}{\sqrt{N}} \sum_{i=1}^N [\mathbf{x}_i \otimes (\mathbf{h}_i - \mu_h)' \mathbf{x}_i] \text{vec} \left((\hat{\beta} - \beta)(\hat{\beta} - \beta)' \right) \\ &= \frac{1}{\sqrt{N}} \frac{1}{N} \sum_{i=1}^N [\mathbf{x}_i \otimes (\mathbf{h}_i - \mu_h)' \mathbf{x}_i] \text{vec} \left(\sqrt{N}(\hat{\beta} - \beta) \sqrt{N}(\hat{\beta} - \beta)' \right). \end{aligned} \quad (41)$$

By the law of large numbers the average

$$\frac{1}{N} \sum_{i=1}^N [\mathbf{x}_i \otimes (\mathbf{h}_i - \mu_h)' \mathbf{x}_i] \rightarrow_p E[\mathbf{x} \otimes (\mathbf{h} - \mu_h)' \mathbf{x}].$$

Therefore, (35) is equal to

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' (u_i^2 - \sigma^2) + o_p(1). \quad (42)$$

b. Assume (33) above. Show that

$$E \left[(u_i^2 - \sigma^2)^2 (\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) \right] = E \left[(u_i^2 - \sigma^2)^2 \right] E \left[(\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) \right]. \quad (43)$$

Note that the left-hand side of (43) is the asymptotic variance of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)' (u_i^2 - \sigma^2) \quad (44)$$

(by iterated expectations, each summand in (44) has mean zero).

Expression (43) follows from a straightforward application of iterated expectations, $E[u|\mathbf{x}] = 0$, $E[u^2|\mathbf{x}] = \sigma^2$, and (33). The calculation is

$$\begin{aligned} E \left[(u_i^2 - \sigma^2)^2 (\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) \right] &= E \left[E \left[(u_i^2 - \sigma^2)^2 (\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) \mid \mathbf{x}_i \right] \right] \\ &= E \left[(\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) E \left[(u_i^4 - 2\sigma^2 u_i^2 + \sigma^4) \mid \mathbf{x}_i \right] \right] \\ &= E \left[(\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) (\kappa_4 + 3\sigma^4 - 2\sigma^4 + \sigma^4) \right] \\ &= (\kappa_4 + 2\sigma^4) E \left[(\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) \right] \\ &= E \left[(u_i^2 - \sigma^2)^2 \right] E \left[(\mathbf{h}_i - \mu_h)' (\mathbf{h}_i - \mu_h) \right] \end{aligned}$$

where, as noted above, κ_4 is the fourth-order cumulant of u_i [see (33)].

c. As noted on page 34, we test the null hypothesis of homoscedasticity in the model $y = \beta_0 + \mathbf{x}\beta + u$ by testing the null hypothesis that $\delta = \mathbf{0}$ in the model

$$u_i^2 = \delta_0 + \mathbf{h}(\mathbf{x}_i)\delta + v_i. \quad (45)$$

To deal with (45), as remarked above, we replace the unobserved u_i^2 with $\hat{u}_i^2$, the squared residuals from the OLS regression in (28). That is, we perform the regression given by (6.38) in W:

$$\hat{u}_i^2 \quad \text{on} \quad 1, \mathbf{h}_i, \quad i = 1, \dots, N. \quad (6.38)$$

Instead of a standard Wald test or a standard F -test, this problem considers the Lagrange multiplier test of the null hypothesis of homoscedasticity. The Lagrange multiplier test (also called the score test) is described in Section 4.2.4 in W. Next semester I will discuss it in more detail.

The Lagrange multiplier test of the null hypothesis given by (29) uses the statistic given by

$$NR^2 \text{ from the regression (6.38).}$$

We need to show this NR^2 statistic has in the limit as $N \rightarrow \infty$ a chi-square distribution with Q degrees of freedom, identical to the limiting distribution of the Wald statistic used to test the same null hypothesis. The null hypothesis is rejected if this test statistic exceeds the $100(1 - \alpha)$ quantile of the chi-square distribution with Q degrees of freedom.

Step (i). In part a, we showed (6.39) in W,

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{h}_i'(\hat{u}_i^2 - \hat{\sigma}^2) = \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(u_i^2 - \sigma^2) + o_p(1). \quad (6.39)$$

As we have noted,

$$\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2) = \sum_{i=1}^N \hat{u}_i^2 - \sum_{j=1}^N \hat{u}_j^2 = 0,$$

and thus we can replace $\mathbf{h}_i'$ in the left-hand summation of (6.39) by $(\mathbf{h}_i - \bar{\mathbf{h}})'$, where

$$\bar{\mathbf{h}} = \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i,$$

yielding

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})'(\hat{u}_i^2 - \hat{\sigma}^2) = \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(u_i^2 - \sigma^2) + o_p(1). \quad (6.39')$$

Consider the sum

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \mu_h)'(u_i^2 - \sigma^2). \quad (46)$$

Under the null hypothesis (29), this expression is $1/\sqrt{N}$ times a sum of iid random vectors with mean $\mathbf{0}$ (use iterated expectations on the summands). By the central limit theorem, (46) has a limiting normal distribution with mean $\mathbf{0}$ and covariance matrix

$$E[(u_i^2 - \sigma^2)^2(\mathbf{h}_i - \mu_h)'(\mathbf{h}_i - \mu_h)] = E[(u_i^2 - \sigma^2)^2]E[(\mathbf{h}_i - \mu_h)'(\mathbf{h}_i - \mu_h)], \quad (47)$$

where (47) follows from part b of this problem.

Step (ii). Recall that, if the $p \times 1$ vector $\mathbf{y}$ is $N(\mu, \Sigma)$, $(\mathbf{y} - \mu)' \Sigma^{-1} (\mathbf{y} - \mu)$ is chi-square with p degrees of freedom. From the asymptotic normality established in step (i) and the continuous mapping theorem, it follows that the limiting distribution of

$$\frac{\sum_{i=1}^N (\mathbf{h}_i - \mu_h)(u_i^2 - \sigma^2) [(\mathbf{h} - \mu_h)'(\mathbf{h} - \mu_h)]^{-1} \sum_{j=1}^N (\mathbf{h}_j - \mu_h)'(u_j^2 - \sigma^2)}{NE[(u^2 - \sigma^2)^2]} \quad (48)$$

under the null hypothesis is chi-square with Q degrees of freedom. Moreover, by (6.39'), the law of large numbers, Slutsky's theorem, and the continuous mapping theorem,

$$\frac{\sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}) \left[\frac{1}{N} \sum_{j=1}^N (\mathbf{h}_j - \bar{\mathbf{h}})'(\mathbf{h}_j - \bar{\mathbf{h}}) \right]^{-1} \sum_{k=1}^N (\mathbf{h}_k - \bar{\mathbf{h}})'(\hat{u}_k^2 - \hat{\sigma})}{\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma})^2} \quad (49)$$

has the same limiting distribution as (48) under the null hypothesis, chi-square with Q degrees of freedom.

Step (iii). Finally, we need to show that (49) is equal to NR^2 from the regression (6.38). First, the sum of squared regression residuals is

$$\sum_{i=1}^N \left(\hat{u}_i^2 - \hat{\delta}_0 - \mathbf{h}_i \hat{\delta} \right)^2 = \sum_{i=1}^N \hat{u}_i^4 - \sum_{i=1}^N \hat{u}_i^2 (\hat{\delta}_0 + \mathbf{h}_i \hat{\delta}), \quad (50)$$

which follows from

$$\sum_{i=1}^N (\hat{u}_i^2 - \hat{\delta}_0 - \mathbf{h}_i \hat{\delta}) = 0 \quad \text{and} \quad \sum_{i=1}^N \mathbf{h}_i (\hat{u}_i^2 - \hat{\delta}_0 - \mathbf{h}_i \hat{\delta}) = \mathbf{0}'$$

(the residuals sum to zero and are orthogonal to the explanatory variables). Second, the total sum of squares of the response variable is

$$\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2.$$

Therefore, NR^2 from the regression (6.38) is

$$NR^2 = N - \frac{N \sum_{i=1}^N \hat{u}_i^4 - N \sum_{i=1}^N \hat{u}_i^2 (\hat{\delta}_0 + \mathbf{h}_i \hat{\delta})}{\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2} = \frac{-N^2 \hat{\sigma}^4 - N \sum_{i=1}^N \hat{u}_i^2 (\hat{\delta}_0 + \mathbf{h}_i \hat{\delta})}{\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2} \quad (51)$$

Next, substitute in (51) for $\hat{\delta}_0$, using

$$\hat{\delta}_0 = \frac{1}{N} \sum_{i=1}^N \hat{u}_i^2 - \bar{\mathbf{h}} \hat{\delta} = \hat{\sigma}^2 - \bar{\mathbf{h}} \hat{\delta}.$$

This follows from the sample analog of (2.40) in W. With the substitution, we obtain

$$NR^2 = \frac{N \sum_{i=1}^N \hat{u}_i^2 (\mathbf{h}_i - \bar{\mathbf{h}}) \hat{\delta}}{\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2} = \frac{N \sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2) (\mathbf{h}_i - \bar{\mathbf{h}}) \hat{\delta}}{\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2}. \quad (52)$$

The least squares estimate $\hat{\delta}$ is given by the sample analog of (2.39) in W. It is

$$\begin{aligned} \hat{\delta} &= [\widehat{Var}(\mathbf{h})]^{-1} \widehat{Cov}(\mathbf{h}, \hat{u}^2) = \left(\sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})' (\mathbf{h}_i - \bar{\mathbf{h}}) \right)^{-1} \sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})' \hat{u}_i^2 \\ &= \left(\sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})' (\mathbf{h}_i - \bar{\mathbf{h}}) \right)^{-1} \sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})' (\hat{u}_i^2 - \hat{\sigma}^2). \end{aligned} \quad (53)$$

Substitution of this into (52) yields (49), completing the derivation.

d. Derive a test of heteroscedasticity without assumption (33),

$$E[u_i^4 | \mathbf{x}_i] = \kappa_4 + 3\sigma^4. \quad (32)$$

That is, in the model $y = \beta_0 + \mathbf{x}\beta + u$, with $E[u|\mathbf{x}] = 0$ and $E[u^2|\mathbf{x}] = \sigma^2$ assumed, we cannot assume that the fourth conditional moment of u is constant.

Wooldridge states that there is a Lagrange multiplier type of test of the null hypothesis (29). The statistic for the test is $N - SSR_{resids}$ from the regression *without an intercept* of

$$1 \quad \text{on} \quad (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2), \quad i = 1, \dots, N, \quad (54)$$

where, as above, the $\hat{u}_i$ are the residuals from OLS for $y = \beta_0 + \mathbf{x}\beta + u$ and $\hat{\sigma}^2$ is defined by (34).

Let's show that, under the null hypothesis, this statistic asymptotically is chi-square with Q degrees of freedom.

Step (i). The model for the regression (54) is

$$1 = (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2)\gamma + \text{error}.$$

The residuals do not add to zero because there is no intercept term. The OLS estimator of the parameter vector γ is

$$\hat{\gamma} = \left[\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2 (\mathbf{h}_i - \bar{\mathbf{h}})' (\mathbf{h}_i - \bar{\mathbf{h}}) \right]^{-1} \sum_{j=1}^N (\mathbf{h}_j - \bar{\mathbf{h}})' (\hat{u}_j^2 - \hat{\sigma}^2) \cdot 1. \quad (55)$$

The residuals are orthogonal to the explanatory variables in the regression, so that

$$\sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)(\mathbf{h}_i - \bar{\mathbf{h}})' [1 - (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2)\hat{\gamma}] = \mathbf{0}. \quad (56)$$

Step (ii). N minus the sum of squared residuals from the regression (54) is

$$\begin{aligned} N - \sum_{i=1}^N [1 - (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2)\hat{\gamma}]^2 \\ &= N - \sum_{i=1}^N [1 - (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2)\hat{\gamma}] \\ &= \sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2)\hat{\gamma} \\ &= \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbf{h}_i - \bar{\mathbf{h}})(\hat{u}_i^2 - \hat{\sigma}^2) \left[\frac{1}{N} \sum_{j=1}^N (\hat{u}_j^2 - \hat{\sigma}^2)^2 (\mathbf{h}_j - \bar{\mathbf{h}})' (\mathbf{h}_j - \bar{\mathbf{h}}) \right]^{-1} \frac{1}{\sqrt{N}} \sum_{k=1}^N (\mathbf{h}_k - \bar{\mathbf{h}})(\hat{u}_k^2 - \hat{\sigma}^2), \end{aligned} \quad (57)$$

where the second line follows from (56) and the last line follows from (55).

Step (iii). The last line of (57) has an asymptotic chi-square distribution with Q degrees of freedom, because

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\hat{u}_i^2 - \hat{\sigma}^2)^2 (\mathbf{h}_i - \bar{\mathbf{h}})' (\mathbf{h}_i - \bar{\mathbf{h}})$$

is a consistent estimator of the asymptotic covariance matrix of (6.39'). This complete the proof of part d.

Appendix. A useful tool from linear algebra.

Kronecker product. Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{B}$ a $p \times q$ matrix. The Kronecker production of $\mathbf{A}$ and $\mathbf{B}$ is the $mp \times nq$ matrix

$$\begin{pmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ & \ddots & \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{pmatrix}, \quad \text{denoted } \mathbf{A} \otimes \mathbf{B}.$$

It is an arrangement of all possible products of one element from $\mathbf{A}$ and one element from $\mathbf{B}$. There are two commands in R for the Kronecker product of $\mathbf{A}$ and $\mathbf{B}$, `Kronecker(A,B)` and `A%x%B`.

The vec operator. The `vec` operator applied to a matrix is a stacking of the columns of the matrix, from the first to the last, producing a column vector. In this notes, we use the following result for a triple matrix product,

$$vec(\mathbf{DEF}) = (\mathbf{F}' \otimes \mathbf{D})vec(\mathbf{E}).$$